**Diagnostic Report generated by gemini-2.5-pro**

# Diagnostic Analysis of Run_29

## 1. Summary of Key Deviations

The analysis of Run_29 reveals a moderate, system-wide loss of flow, with a particularly unusual signature in the bypass loop. The majority of the system's sensors show no deviation at all.

- **Flow_Bypass (Moderate):** The most significant anomaly is a large **negative offset** on the bypass flow, which reads approximately **-18% to -20%** lower than healthy. This is the inverse of what is seen in blockage scenarios and is a key diagnostic clue.
- **Flow_Pump (Moderate):** The pump flow shows a corresponding moderate **negative offset** of approximately **-10%**. This indicates a system-wide loss of flow originating at or before the pump.
- **All Other Signals (Nominal):** Crucially, **all pressure sensors** (`p_Pump`, `p_FMU`, `p_Combustion`, `p_Shut`) and **all downstream flow sensors** (`Flow_Engine1-4`, `Flow_PRV`) show **no significant deviations**.

## 2. Root-Cause Hypotheses

| Rank | Suspected Component | Fault Type | Supporting Evidence |
|---|---|---|---|
| 1 | **Bypass Valve** | **Stuck Significantly Open** | This is the most compelling hypothesis. If the bypass valve is stuck open, it diverts a large portion of the pump's output directly back to the inlet *before* it can be properly utilized or measured by the main flow sensors. This perfectly explains the large negative offsets on both **Flow_Bypass** (-20%) and **Flow_Pump** (-10%). The system is producing flow, but it's being immediately lost in an internal leak/loop, starving the rest of the system. |

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|---|---|---|---|
| 2 | **Flow_Pump & Flow_Bypass Sensors** | **Dual Sensor Failure (Negative Offset)** | It's possible both sensors have failed with a negative offset. However, the fact that both are negative and the rest of the system is healthy points more strongly to a single physical cause that would affect both readings in this specific way. A dual sensor failure is less probable than a single component failure. |
| 3 | **Main Fuel Pump** | **Moderate Internal Leakage / Degradation** | A degraded pump would explain the `Flow_Pump` drop of -10%. However, it doesn't as elegantly explain why the `Flow_Bypass` reading is so disproportionately low (-20%). A stuck-open valve provides a better explanation for the specific relationship between these two signals. |
| 4 | **Boost-Pressurised Tank / Inlet Line** | **Moderate Blockage** | A moderate blockage before the pump would starve it of fuel, causing a drop in `Flow_Pump` and `Flow_Bypass`. This is a plausible alternative for the physical fault, but again, doesn't explain the particularly severe drop in the bypass reading as well as a stuck valve does. |
| 5 | **System Contamination** | **Debris causing valve to stick** | This is the likely underlying cause of the stuck valve (Hypothesis 1). A piece of debris could be preventing the bypass valve from closing correctly. |

## 3. Most Probable Hypothesis

The most probable root cause is that the **Bypass Valve is Stuck Significantly Open.**

**Reasoning:** This single mechanical failure provides the most complete and parsimonious explanation for the unique data signature. When the bypass valve sticks open, a large portion of the pump's output is immediately "leaked" back to the pump inlet. This has two key effects seen in the data: 1) The overall flow available to the rest of the system drops, causing the `Flow_Pump` reading to be low (-10%). 2) The flow through the *measured and controlled* part of the bypass circuit, where the `Flow_Bypass` sensor is, also drops dramatically (-20%) because most of the diverted flow is going through the unintended leak path. This one fault elegantly explains both major negative flow deviations while also being consistent with the complete lack of deviation in any other downstream pressure or flow sensor.